

```
# Write a script that declares variables of each type (int, float, str, bool)

number = 10          # int
decimal = 5.5        # float
name = "Alex"        # string
is_student = True    # boolean

#performs arithmetic operations
add = number + decimal
subtract = number - decimal
multiply = number * decimal
division = number / decimal

# prints formatted output using f-strings.
print(f"Name: {name}")
print(f"Is student: {is_student}")
print(f"Addition: {add}")
print(f"Subtraction: {subtract}")
print(f"Multiplication: {multiply}")
print(f"Division: {division}")
```

```
Name: Alex
Is student: True
Addition: 15.5
Subtraction: 4.5
Multiplication: 55.0
Division: 1.8181818181818181
```

```
# Implement a function that accepts two numbers and returns their:
# sum, difference, product, and quotient as a tuple
#call it with at least three different input pairs.
def calculate(a, b):
    return (a + b, a - b, a * b, a / b)

for numbers in [(10, 5), (20, 4), (8, 2)]:
    total, difference, product, quotient = calculate(numbers[0], numbers[1])

    print(f"Numbers: {numbers[0]} and {numbers[1]}")
    print(f"Sum = {total}")
    print(f"Difference = {difference}")
    print(f"Product = {product}")
    print(f"Quotient = {quotient}")
    print()
```

```
Numbers: 10 and 5
Sum = 15
Difference = 5
Product = 50
Quotient = 2.0
```

```
Numbers: 20 and 4
Sum = 24
```

```
Difference = 16
Product = 80
Quotient = 5.0
```

```
Numbers: 8 and 2
Sum = 10
Difference = 6
Product = 16
Quotient = 4.0
```

```
#Use a for loop with range() to generate a multiplication table (1-10)
#print it in a formatted grid.
for number in range(1, 11):
    for multiplier in range(1, 11):
        result = number * multiplier
        print(f"{result:4}", end="")
    print()
```

```
1  2  3  4  5  6  7  8  9 10
2  4  6  8 10 12 14 16 18 20
3  6  9 12 15 18 21 24 27 30
4  8 12 16 20 24 28 32 36 40
5 10 15 20 25 30 35 40 45 50
6 12 18 24 30 36 42 48 54 60
7 14 21 28 35 42 49 56 63 70
8 16 24 32 40 48 56 64 72 80
9 18 27 36 45 54 63 72 81 90
10 20 30 40 50 60 70 80 90 100
```

```
# Write a temperature converter that reads a value and unit (C/F)
# uses if/elif/else to convert and display the result.
```

```
temperature = float(input("Enter temperature: "))
unit = input("Enter unit (C/F): ").upper()
```

```
if unit == "C":
    result = (temperature * 9/5) + 32
    print(f"{temperature}C is {result}F")
```

```
elif unit == "F":
    result = (temperature - 32) * 5/9
    print(f"{temperature}F is {result}C")
```

```
else:
    print("Invalid unit")
```

```
Enter temperature: 45
Enter unit (C/F): c
45.0C is 113.0F
```

◆ Gemini

```
students = [
    {"name": "Jacob", "scores": [80, 90, 70]},
```

```
    {"name": "James", "scores": [60, 75, 85]},
    {"name": "Josh", "scores": [95, 90, 100]}
]

# Calculate averages

for student in students:
    total = sum(student["scores"])
    total = builtins.sum(student["scores"])
    average = total / len(student["scores"])
    student["average"] = average

students.sort(key=lambda x: x["average"], reverse=True)

print("Leaderboard")

for student in students:
    print(f"{student['name']} - Average: {student['average']}")
```

```
Leaderboard
Josh - Average: 95.0
Jacob - Average: 80.0
James - Average: 73.33333333333333
```

```
#Write a list comprehension that filters even numbers from a range
#squares them
#then write an equivalent dict comprehension mapping each word in a sentence

numbers = range(1, 11)

even_squares = [num ** 2 for num in numbers if num % 2 == 0]

print(even_squares)

# dictionary version
sentence = "I like to play basketball"

words = sentence.split()

word_lengths = {word: len(word) for word in words}

print(word_lengths)
```

```
[4, 16, 36, 64, 100]
{'I': 1, 'like': 4, 'to': 2, 'play': 4, 'basketball': 10}
```

```
import csv
```

```
totals = []
```

```
count = 0

# Read CSV

with open("grades.csv", "r") as file:
    reader = csv.DictReader(file)

    for row in reader:
        count += 1

        if not totals:
            totals = [0] * len(row)

        for index, value in enumerate(row.values()):
            totals[index] += int(value)

# Calculate averages

averages = []

for total in totals:
    averages.append(total / count)

# Write summary CSV

with open("summary.csv", "w", newline="") as file:
    writer = csv.writer(file)

    writer.writerow(["Column Average"])

    for average in averages:
        writer.writerow([average])

print("Summary created")
```

```
-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_772/999558719.py in <cell line: 0>()
      8 # Read CSV
      9
----> 10 with open("grades.csv", "r") as file:
      11     reader = csv.DictReader(file)
      12

FileNotFoundError: [Errno 2] No such file or directory: 'grades.csv'
```

Next steps: [Explain error](#)

